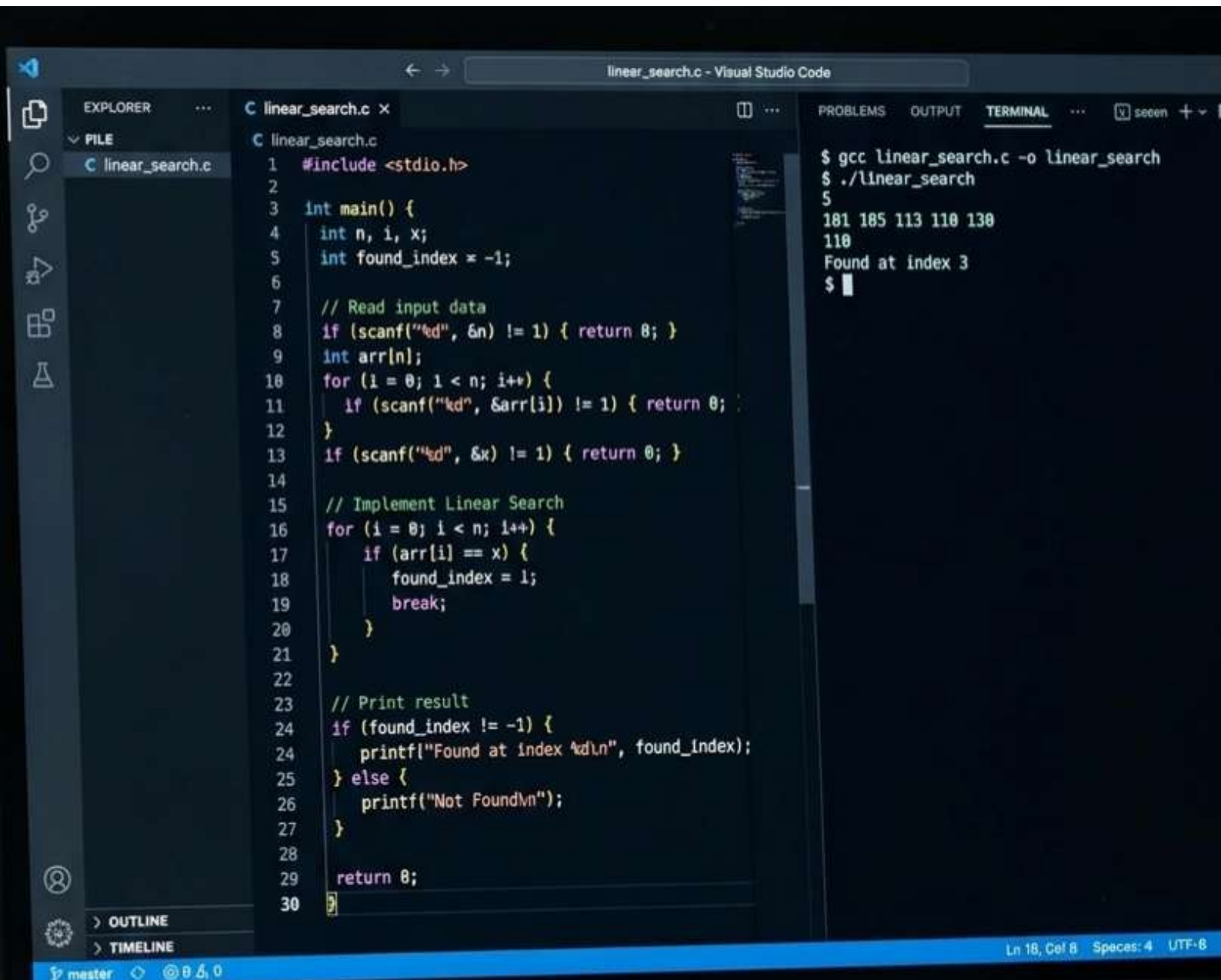


Week - 2 : ModuleDate - 17/06/2026Solution - 1

```
#include <stdio.h>
int main()
{
    int n, x;
    int found_index = -1;
    if (scanf("%d", &n) != 1) return 0;
    int roll_numbers[n];
    for (int i = 0; i < n; i++)
    {
        if (scanf("%d", &roll_numbers[i]) != 1)
            return 0;
    }
    if (scanf("%d", &x) != 1) return 0;
    for (int i = 0; i < n; i++)
    {
        if (roll_numbers[i] == x) {
            found_index = i;
            break;
        }
    }
}
```



```
linear_search.c - Visual Studio Code

EXPLORER
FILE
C linear_search.c

C linear_search.c
1 #include <stdio.h>
2
3 int main() {
4     int n, i, x;
5     int found_index = -1;
6
7     // Read input data
8     if (scanf("%d", &n) != 1) { return 0; }
9     int arr[n];
10    for (i = 0; i < n; i++) {
11        if (scanf("%d", &arr[i]) != 1) { return 0; }
12    }
13    if (scanf("%d", &x) != 1) { return 0; }
14
15    // Implement Linear Search
16    for (i = 0; i < n; i++) {
17        if (arr[i] == x) {
18            found_index = i;
19            break;
20        }
21    }
22
23    // Print result
24    if (found_index != -1) {
25        printf("Found at index %d\n", found_index);
26    } else {
27        printf("Not Found\n");
28    }
29    return 0;
30 }

PROBLEMS OUTPUT TERMINAL
$ gcc linear_search.c -o linear_search
$ ./linear_search
5
181 185 113 110 130
110
Found at index 3
$
```

Ln 18, Col 8 Specs: 4 UTF-8

Solution - 2 :-

```
#include <stdio.h>
int main()
{
    int n, target;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    for (int i = 0; i < n; i++)
    {
        if (scanf("%d", arr[i]) != 1) return 0;
    }
    if (scanf("%d", &target) != 1)
    {
        int low = 0;
        int high = n - 1;
        int found = 0;
        while (low <= high) {
            int mid = low + (high - low) / 2;
            if (arr[mid] == target) {
                found = 1;
                break;
            } else if (arr[mid] < target) {
                low = mid + 1;
            } else {
                high = mid - 1;
            }
        }
        if (found) {
            printf("Product Available\n");
        } else {
            printf("Product Not Available\n");
        }
        return 0;
    }
}
```


C search_product.c ×

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TERMINAL

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c > C search_product.c > main

```
1  #include <stdio.h>
2
3  // Function for Binary Search
4  int binarySearch(int arr[], int n, int target) {
5      int low = 0;
6      int high = n - 1;
7
8      while (low <= high) {
9          int mid = low + (high - low) / 2;
10
11         if (arr[mid] == target)
12             return 1; // Product found
13
14         if (arr[mid] < target)
15             low = mid + 1;
16         else
17             high = mid - 1;
18     }
19
20     return 0; // Product not found
21 }
22
23 int main() {
24     int n, target;
25
26     // Reading Input Format
27     if (scanf("%d", &n) != 1) return 0;
28
29     int product_ids[n];
30     for (int i = 0; i < n; i++) {
31         if (scanf("%d", &product_ids[i]) != 1) return 0;
32     }
33
34     if (scanf("%d", &target) != 1) return 0;
35
36     // Binary search logic
37     if (binarySearch(product_ids, n, target)) {
38         printf("Product Available\n");
39     } else {
40         printf("Product Not Available\n");
41     }
42
43     return 0;
44 }
```

```
~/Users/anakops/search/se
gcc search_product.c -o s
$ ./search
Example Input:
6
10 20 30 40 50 60
40
Example Output:
Product Available
$
```

Solution - 3 :-

```
#include <stdio.h>
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    for (i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) return 0;
    }
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    for (int i = 0; i < n; i++) {
        printf("%d", arr[i]);
        if (i < n - 1) {
            printf(" ");
        }
    }
    printf("\n");
    return 0;
}
```

EXP... ...

C bubble_sort.c X

FILE

C bubble_...

C bubble_sort.c > ...

```
1  #include <stdio.h>
2  int main()
3  {
4      int n, i, j, temp;
5      scanf("%d", &n);
6
7      for(i=0; i<n-1; i++) {
8          for(j=0; j<n-i-1; j++) {
10             if(arr[j] > arr[j+1]) {
11                 temp = arr[j];
12                 arr[j] = arr[j+1];
14                 arr[j+1] = temp;
15             }
16         }
17     }
19
20     for(i=0; i<n; i++) {
21         printf("%d ", arr[i]);
23     }
24     printf("\n");
25
26     return 0;
27 }
28
```

PROBLEMS TERMINAL TERMINAL

shell + - [] [] ^ x

```
~/projects/bubble_sort$ gcc bubble_sort.c -o bubble_sort
~/projects/bubble_sort$ ./bubble_sort
Input:
5
89 45 78 12 99
Output:
12 45 78 89 99
~/projects/bubble_sort$
```


Solution - 4 -

```
#include <stdio.h>
int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) return 0;
    }
    for (int i = 1; i < n; i++) {
        int Key = arr[i];
        int j = i - 1;
        while (j > 0 && arr[j] > Key) {
            arr[j+1] = arr[j];
            j = j - 1;
        }
        arr[j+1] = Key;
    }
    for (int i = 0; i < n; i++) {
        printf("%d", arr[i]);
        if (i < n-1) {
            printf(" ");
        }
    }
    printf("\n");
    return 0;
}
```


Solution-5 :-

```
#include <stdio.h>
int main()
{
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int arr[n];
    for (int i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) return 0;
    }
    for (int i = 0; i < n - 1; i++) {
        int min_idx = i;
        for (int j = i + 1; j < n; j++) {
            if (arr[j] < arr[min_idx])
                min_idx = j;
        }
        int temp = arr[min_idx];
        arr[min_idx] = arr[i];
        arr[i] = temp;
    }
    for (int i = 0; i < n; i++) {
        printf("%d", arr[i]);
        if (i < n - 1) {
            printf(" ");
        }
    }
    printf("\n");
    return 0;
}
```



C rank_players.c ×



C rank_players.c > ...

```

1  #include <stdio.h>
2
3  int main() {
4      int n, i, j, min_idx, temp;
5
6      if (scanf("%d", &n) != 1) return 0;
7
8      int arr[n];
9
10     for (i = 0; i < n; i++) {
11         if (scanf("%d", &arr[i]) != 1) return 0;
12     }
13
14     // Selection Sort
15     for (i = 0; i < n - 1; i++) {
16         min_idx = i;
17         for (j = i + 1; j < n; j++) {
18             if (arr[j] < arr[min_idx]) {
19                 min_idx = j;
20             }
21         }
22         temp = arr[min_idx];
23         arr[min_idx] = arr[i];
24         arr[i] = temp;
25     }
26
27     for (i = 0; i < n; i++) {
28         printf("%d", arr[i]);
29         if (i < n - 1) {
30             printf(" ");
31         }
32     }
33     printf("\n");
34
35     return 0;
36 }
```



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

One Dark Pi + ▾ □ 🗑 ⤴ ×

```

C:\Users\User\Code>gcc rank_players.c -o rank_players
C:\Users\User\Code>./rank_players
Input: 5
70 20 90 10 40
Output: 10 20 40 70 90

C:\Users\User\Code>|
```